4.17 Demonstrate an understanding of the idea of conservation of energy in various energy transfers

4.18 **Carry out calculations on work done to show the dependence of braking distance for a vehicle on initial velocity squared (work done to bring a vehicle to rest equals its initial kinetic energy)**

Topic 5

Nuclear fission and nuclear fusion

5.1 Describe the structure of nuclei of isotopes using the terms atomic (proton) number and mass (nucleon) number and using symbols in the format $^{14}_{6}C$

5.2 Explain how atoms may gain or lose electrons to form ions

5.3 Recall that alpha and beta particles and gamma rays are ionising radiations emitted from unstable nuclei in a random process

5.4 Recall that an alpha particle is equivalent to a helium nucleus, a beta particle is an electron emitted from the nucleus and a gamma ray is electromagnetic radiation

5.5 Compare alpha, beta and gamma radiations in terms of their abilities to penetrate and ionise

5.6 Demonstrate an understanding that nuclear reactions can be a source of energy, including fission, fusion and radioactive decay

5.7 Explain how the fission of U-235 produces two daughter nuclei and two or more neutrons, accompanied by a release of energy

5.8 Explain the principle of a controlled nuclear chain reaction

5.9 Explain how the chain reaction is controlled in a nuclear reactor including the action of moderators and control rods

5.10 Describe how thermal (heat) energy from the chain reaction is converted into electrical energy in a nuclear power station

5.11 Recall that the products of nuclear fission are radioactive

5.12 Describe nuclear fusion as the creation of larger nuclei from smaller nuclei, accompanied by a release of energy and recognise fusion as the energy source for stars